VexarDrive Fleet Analytics

Technical Report — Driver Behaviour & Vehicle Health Analysis

1. Executive Summary

This report documents the analysis of the supplied VexarDrive fleet dataset. The objective is to identify risky driver behaviour and vehicles that warrant maintenance attention based on irregular sensor signatures and vehicle-use indicators.

The analysis covers 30 drivers, 30 vehicles, 450 trips and 12,987 telemetry observations. Seven drivers were classified as Higher Risk, 12 as Moderate Risk and 11 as Lower Risk. Seven vehicles were classified as Higher Attention, eight as Monitor and 15 as Lower Attention.

2. Dataset and Data Quality

Dataset	Rows	Purpose
Drivers	30	Driver attributes
Vehicles	30	Vehicle, odometer and service information
Trips	450	Trip distance, duration and speed
Telemetry	12,987	Sensor, speed and GPS observations

The source tables contained no missing values. Trip_ID was unique in the Trips table, all telemetry Trip_ID values matched Trips, and the final joins introduced no missing values or duplicate columns. Telemetry was enriched using Trip_ID, Driver_ID and Vehicle_ID while preserving the telemetry grain.

3. Driver Behaviour Methodology

Four behaviour indicators were created. Elevated speed was defined as Speed ≥ 45 km/h. Harsh acceleration was Accel_X ≥ 0.5 g, harsh braking was Accel_X ≤ –0.5 g, and aggressive lateral movement was |Accel_Y| ≥ 0.25 g. The thresholds were selected using the observed telemetry distribution; 45 km/h is above the fleet's 95th speed percentile and the acceleration thresholds target extreme observations.

Event counts were converted to rates per 100 telemetry observations to account for differences in telemetry exposure between drivers.

Risk component	Weight
Overspeed rate	35%
Harsh braking rate	25%
Harsh acceleration rate	20%
Aggressive lateral rate	20%

Risk Score = 0.35×Overspeed + 0.25×Braking + 0.20×Acceleration + 0.20×Lateral. Each component is converted to a fleet-relative percentile. Scores below 40 are Lower Risk, 40–70 Moderate Risk, and above 70 Higher Risk.

The score is a relative fleet-risk indicator, not an accident probability.

4. Driver Findings

Rank	Driver	Score	Category
1	D24 — Lakshmi Iyer	91.9	Higher Risk
2	D03 — Vignesh Murugan	90.5	Higher Risk
3	D19 — Senthil Pillai	88.8	Higher Risk
4	D14 — Rajesh Subramaniam	88.6	Higher Risk
5	D06 — Bhavani Raj	88.3	Higher Risk
6	D23 — Kavya Pillai	87.2	Higher Risk
7	D12 — Praveen Murugan	83.4	Higher Risk

Fleet-level event shares were approximately 3.16% elevated-speed observations, 2.99% aggressive lateral observations, 2.11% harsh acceleration observations and 1.99% harsh braking observations.

5. Vehicle Health Methodology

Vehicle status is intentionally described as Maintenance Attention rather than confirmed mechanical failure. Sensor anomalies were identified using fleet-relative 99th-percentile thresholds for absolute sensor measures. This identifies unusual telemetry signatures without claiming that an extreme reading proves a component failure.

Component	Weight	Interpretation
Sensor anomaly rate	50%	Direct telemetry irregularity
Starting odometer	20%	Cumulative usage proxy
Days since service	15%	Maintenance recency
Vehicle age	15%	Age-related wear exposure

Maintenance Attention = 0.50×Sensor + 0.20×Odometer + 0.15×Service + 0.15×Age. Each component is converted to a 0–100 fleet-relative percentile. Scores below 40 are Lower Attention, 40–70 Monitor, and above 70 Higher Attention.

6. Vehicle Findings

Rank	Vehicle	Score	Category
1	V02	94.8	Higher Attention
2	V12	86.7	Higher Attention
3	V19	81.6	Higher Attention
4	V01	78.3	Higher Attention
5	V23	77.9	Higher Attention
6	V10	77.7	Higher Attention
7	V24	73.6	Higher Attention

There were 620 telemetry observations with at least one extreme sensor reading. V02 had the highest observed sensor-anomaly rate at approximately 9.26%. The correlation between starting odometer and maintenance-attention score was approximately 0.772 in the 30-vehicle sample. This is an association, not evidence that mileage alone causes maintenance problems.

7. Assumptions and Limitations

• The thresholds and score weights are analytical assumptions, not externally validated safety or maintenance standards.
• Percentile scores measure relative position within this fleet and should be recalibrated for another fleet.
• Sensor anomalies indicate unusual telemetry signatures, not confirmed mechanical faults.
• Odometer, age and service recency are maintenance-attention indicators, not diagnoses.
• The 30-driver/30-vehicle sample should be validated against a larger historical dataset before operational deployment.

8. Additional Uses of the Dataset

Potential extensions include predictive maintenance, driver coaching, route and operational efficiency analysis, fleet utilization and replacement planning, sensor-quality monitoring, fuel/energy efficiency modelling when relevant data is available, accident-risk modelling after linking historical incidents, and validated insurance/risk segmentation.

9. Conclusion

The analysis provides two transparent prioritization frameworks: a relative Driver Risk Score for identifying behavioural outliers and a Vehicle Maintenance Attention Score for prioritizing vehicles with unusual telemetry and higher wear-related indicators. The accompanying dashboards present the results for operational review, while the documented assumptions make the methodology reproducible and auditable.